SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures **COURSE CODE: CSA03**

Register Number	192511167
Name	GIFTSON KEVIN MATTHEW.Y

COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* (BL2, BL3)

Assessment Tool Weightage: 35%

Questions:2 **EACH QUESTIONS CARRIES:** 10 MARKS **Total Marks:**20 **Duration:** ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

I GIFTSON KEVIN MATTHEW.Y (192511167) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

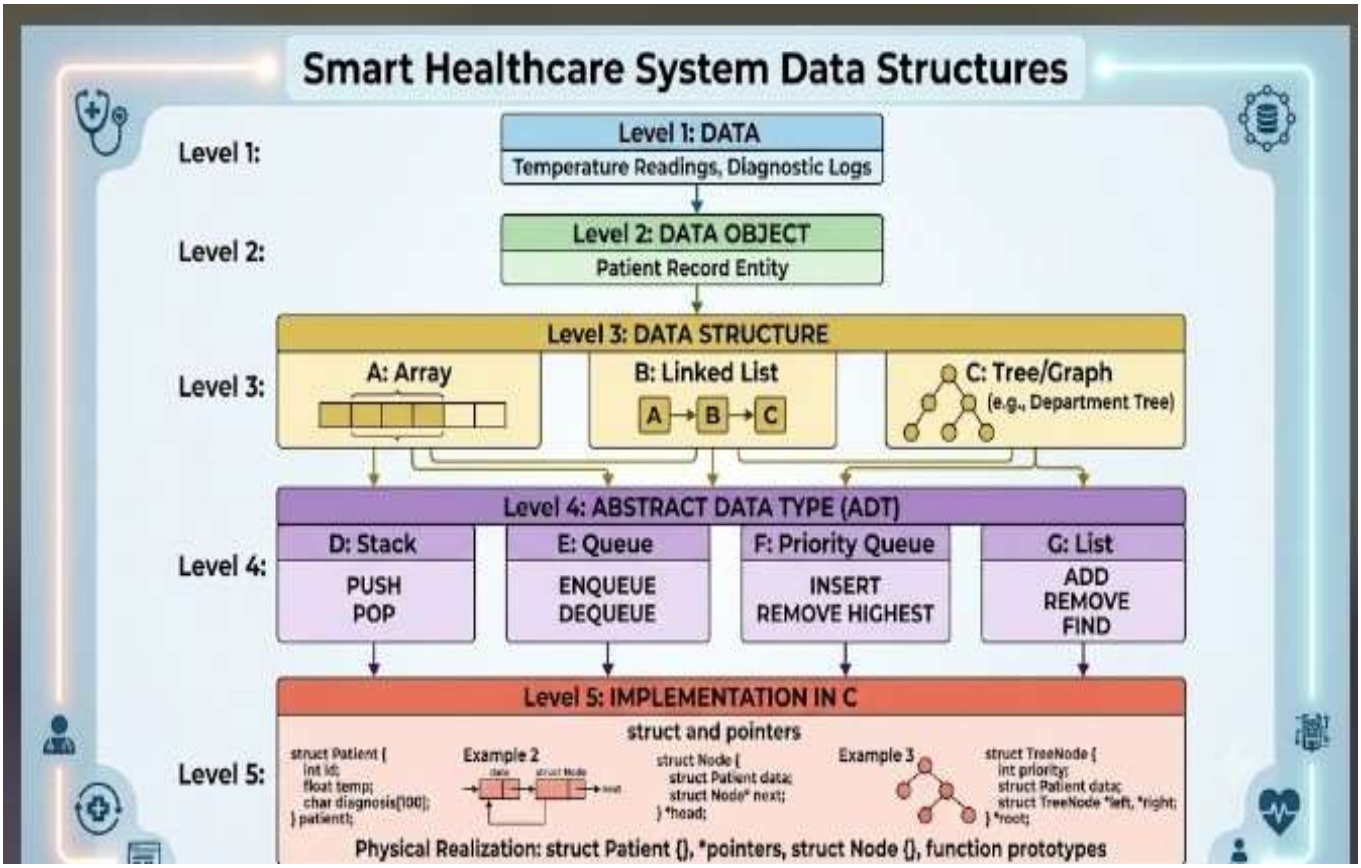
Signature of the student: GIFTSON KEVIN MATTHEW.Y

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Question 1: Smart Healthcare System Concept Map

Focus: Progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C.

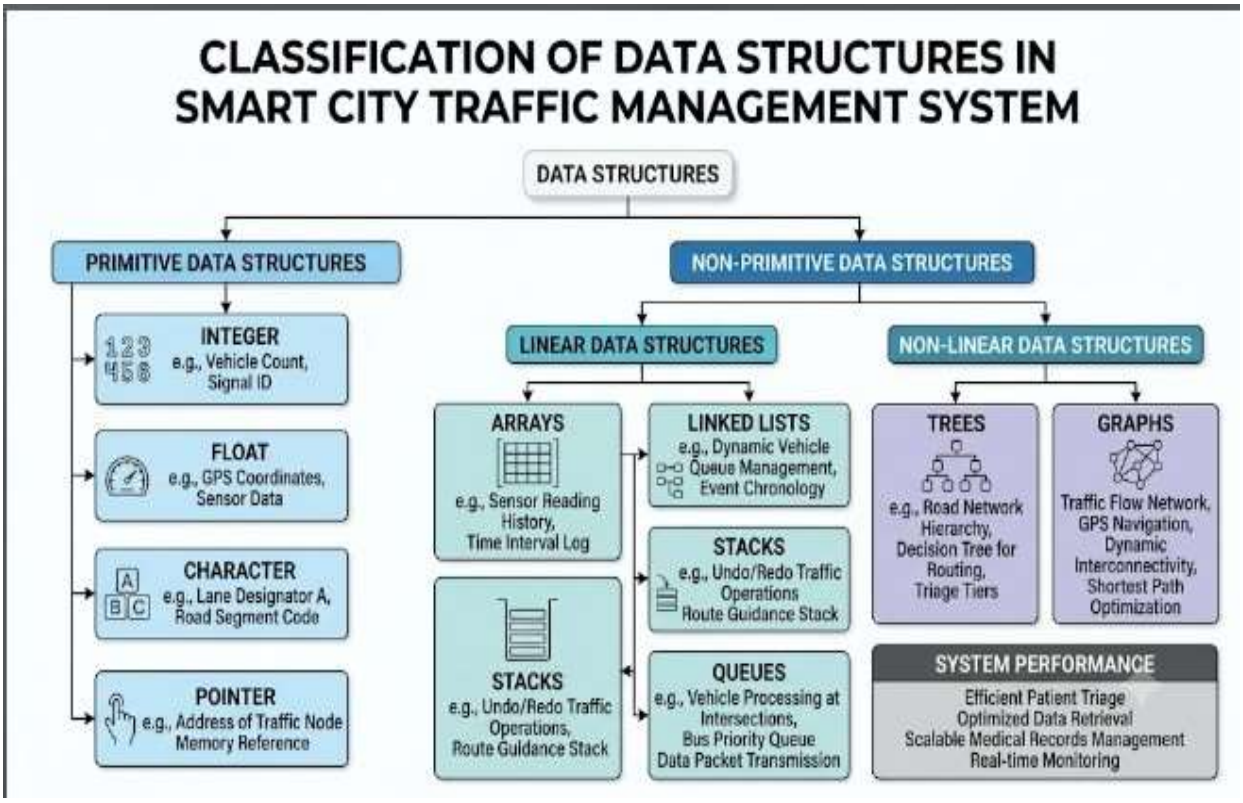


Detailed Content & Justification for Q1:

1. **Data:** Raw, unorganized facts like patient body temperature, heart rate, and raw text diagnostic logs.
2. **Data Object:** Representation of data instances with properties (e.g., a patient record entity containing attributes).
3. **Data Structure:** Organization of data in computer memory (e.g., using struct in C to group patient vitals).
4. **Abstract Data Type (ADT):** Logical definition of data and operations independent of implementation (e.g., Queue ADT for handling emergency patient triage).
5. **Implementation in C:** Concrete realization using pointers, dynamic memory allocation (malloc), and custom structures to maximize memory efficiency and rapid data retrieval for clinical decisions.

Question 2: Smart City Traffic Management System Concept Map

Focus: Classification of Primitive → Non-Primitive → Linear & Non-Linear Data Structures (Arrays, Linked Lists, Trees, Graphs).



Detailed Content & Justification for Q2:

1. **Primitive Data Structures:** Basic built-in types (integers, floats) used to store simple traffic sensor states or speed values.
2. **Non-Primitive Data Structures:** Advanced structures categorized into:
    - **Linear Structures:** Arrays (storing sequential vehicle count logs) and Linked Lists (dynamic vehicle queue updates at traffic lights).
    - **Non-Linear Structures:** Trees (hierarchical route indexing) and Graphs (modeling interconnected road networks, intersections, and shortest-path traffic signal routing).
3. **Functionality & Efficiency Justification:** Graphs enable real-time GPS routing and congestion control, while arrays/lists provide fast sequential access for historical logging.